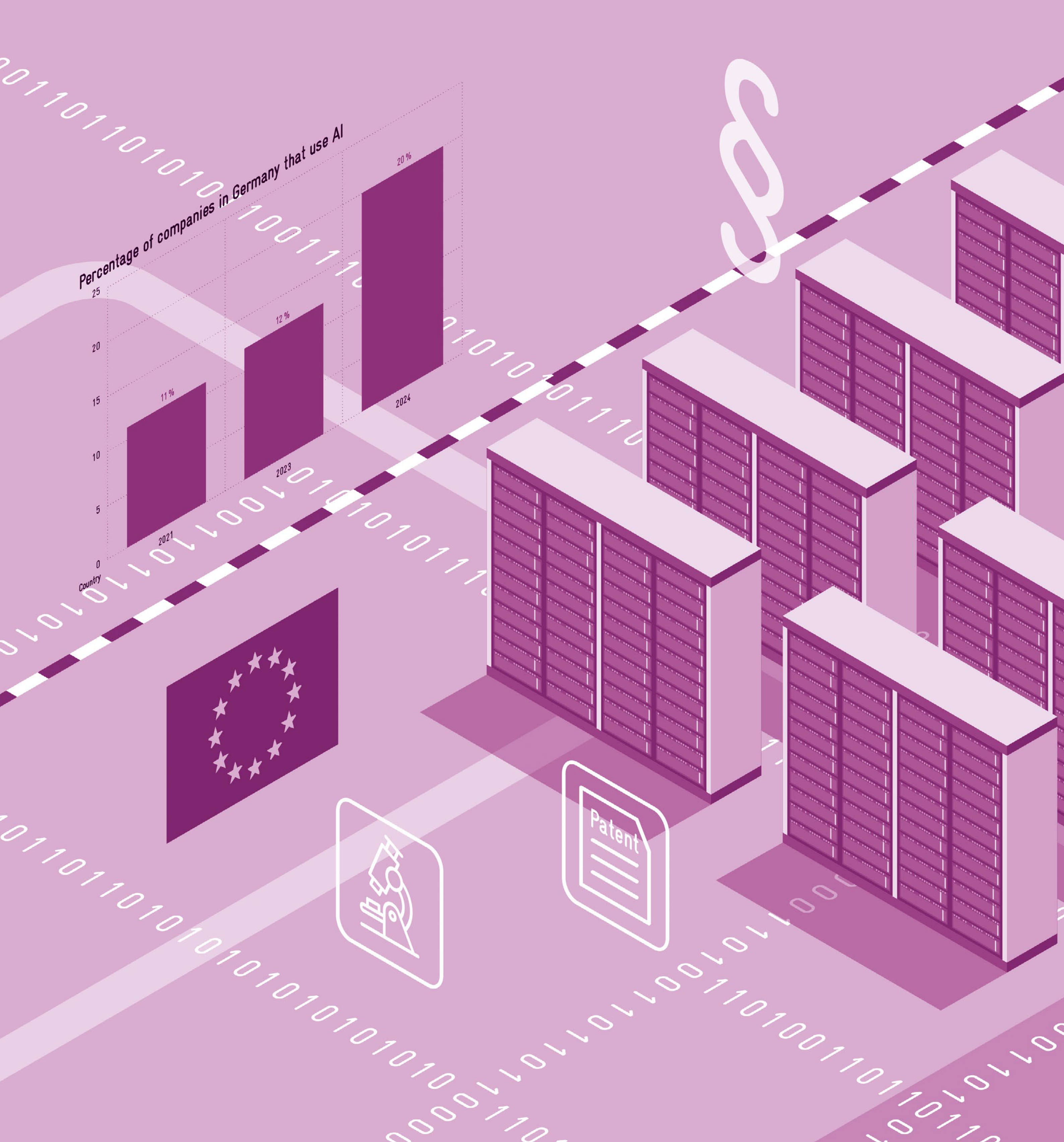
B 3 Development and Application of Artificial Intelligence in Germany and Europe




GIGAFACTORY

B 3 Development and Application of Artificial Intelligence in Germany and Europe

Artificial intelligence (AI) offers enormous potential for innovation and opportunities for economic growth. It promises productivity gains and enables new products, services and business models. AI can help to overcome a wide range of societal challenges – from the early detection of diseases and the optimization of electricity grids in the context of the energy transition to the modernization of government services. At the same time, AI brings challenges in terms of the labour market, international dependencies and ethical aspects of its use. For Germany and the EU, the successful development and application of AI is a key lever for securing the international competitiveness of companies, strengthening digital sovereignty and maintaining prosperity in the long term.

The Federal Government should align its AI strategy with Europe. It is important to avoid technological dependencies and to realize key elements of value creation in Europe. To this end, a powerful AI infrastructure must be established quickly, excellent research and development must be promoted, an innovation-friendly regulatory framework must be created, and the widespread economic application of AI in Germany and the EU must be supported.

B 3-1 Technical Aspects of AI

Diverse Elements in the AI Ecosystem

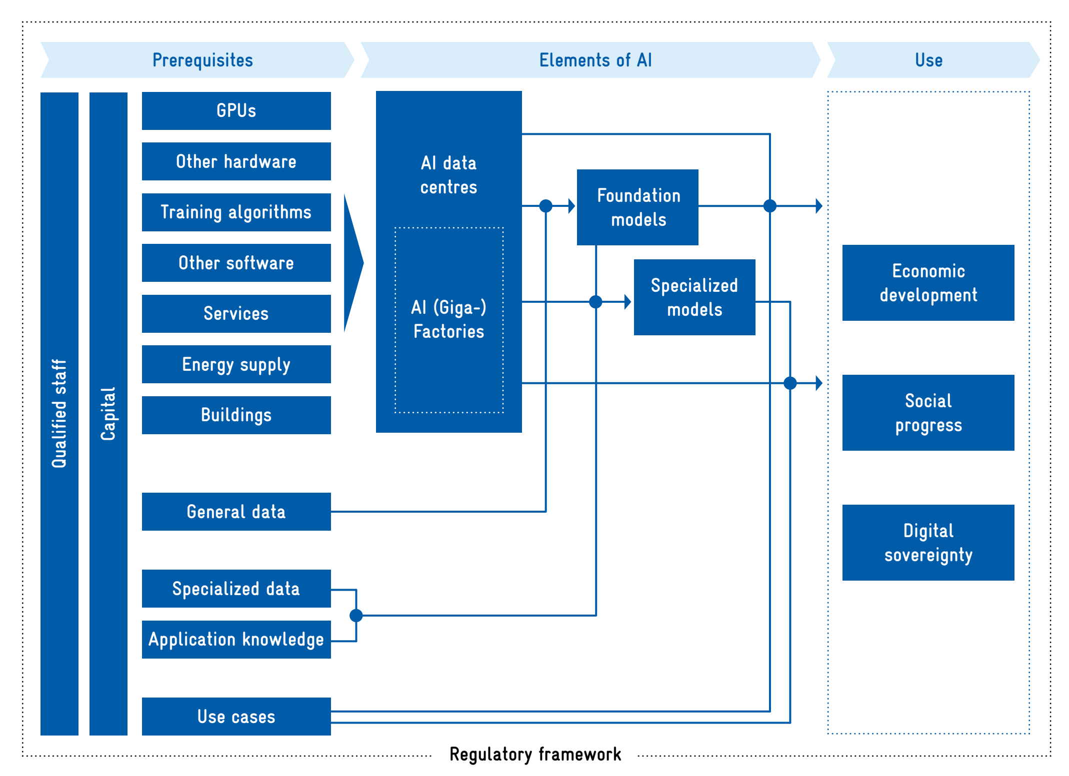
Figure B 3-1 shows the key elements of an AI ecosystem and their interdependencies. The starting point is the prerequisites necessary for setting up and operating AI data centres and, in particular, AI Factories.[387] These include, for example, hardware and software, as well as energy supply and buildings. Other prerequisites include data and application knowledge, which are needed to train AI models. The data flows into foundation models, which provide the underlying technology, and into specialized models developed for specific fields of application. Data and application knowledge are continuously incorporated into the further development of the models. Qualified specialists and capital are overarching key factors that support all fields of the ecosystem. These prerequisites create AI systems that can then be used to generate economic and social benefits. The entire ecosystem is accompanied by a regulatory framework that, among other things, sets rules for the responsible use of AI, clarifies liability issues and strengthens consumer protection.

Rapid Developments in AI

AI is currently undergoing a phase of rapid development, characterized by a sharp increase in investment[388] and the convergence of various strands of development. At the same time, the long-term goal of artificial general intelligence (AGI) is drawing closer – some AI researchers predict that artificial general intelligence will be achieved by around 2040, while others believe it will happen sooner.[389] Various experts emphasize that powerful AI models are now already giving some countries significant economic and military advantages, and that these effects could be significantly amplified once artificial general intelligence is achieved.[390] At the same time, there is a trend towards smaller, specialized models that work more efficiently and cost-effectively than the large foundation models.

Significant developments are also taking place with regard to licensing. Currently, many AI companies offer open models that disclose their parameters and, in some cases, their training data, thus allow-

Fig. B 3-1 Prerequisites and elements of AI development and application



Note: Arrows should be read as 'prerequisite for'. Nodes represent arrows that converge.

Source: Own representation.

ing for customization and royalty-free use. In addition to the early and still influential generative open source models[391] from Western providers such as Meta (Llama), these now include many Chinese companies such as DeepSeek.[392] New standards and formats such as the Model Context Protocol and Safetensors facilitate, among other things, data usage and model storage.[393]

An important trend in terms of applications is physical AI, which enables AI to interact with the real world and is trained accordingly with data from the physical world. The most significant change in application currently comes from AI agents. They enable the operation of systems that can interact autonomously with other systems, retrieve information and perform actions such as sending a message or controlling logistics chains. Companies have already begun to integrate these agents into their business processes.

Graphics processing units (GPUs) are AI chips that were originally developed for demanding computer graphics. They are used for training AI models, processing inference queries and high-performance computing. The GPU market is dominated by the US company Nvidia, with an estimated market share of over 80 percent.[394] Nvidia's proprietary computing platform and interface CUDA, which enables the efficient use of Nvidia GPUs and is not compatible with GPUs from competing providers such as AMD, is also of great importance.[395] Numerous AI tools such as TensorFlow and PyTorch are optimized for CUDA, which favours the dominance of Nvidia's GPUs.

These are countered by Google's TPUs, highly specialized chips (accelerators) that are optimized for extensive AI workloads and play a key role in Google's own services. They have also been made available to other individual companies since 2025.[396] This could create a counterweight to Nvidia's GPU dominance.[397]

Box B 3-2 AI Terminology

Artificial Intelligence: The term artificial intelligence is used to describe processes, algorithms and technological solutions that make it possible to transfer complex cognitive processes previously carried out by humans to learning machines and software.[398]

Generative AI: Generative AI is a form of AI that is used to generate or edit content such as text, images, video, audio and computer code. Unimodal AI systems can only work with one type of data (e.g. only text or only images), while multimodal AI systems can process different types of data and, if necessary, also output different types of data.

Physical AI: Physical AI is an AI system that is not limited to the digital space, but perceives the physical world via sensors, learns from it and interacts with it.[399] Applications include robotics and autonomous driving.

AI Agents: An AI agent is a system that independently collects and processes information and translates it into actions. An AI agent can, for example, record stock levels, forecast demand and, as a result, trigger orders.[400]

Machine Learning: Machine learning uses learning algorithms and data to train complex AI models, which are then applied to new data of the same type.[401]

Parameters: Parameters are numerical values determined by machine learning models during training.[402] The number of parameters in a model influences the model's capability.

Artificial General Intelligence: Artificial general intelligence is a hypothetical form of AI that has the ability to master or learn any cognitive task that a human can perform, at a level that is at least equal to that of a human. Unlike current AI systems, which are usually specialized for specific tasks, artificial general intelligence systems have comprehensive general knowledge and can transfer knowledge.[403]

Foundation Models: Foundation models are generative AI models that are trained on a broad, general database. They form the basis for the development of specific applications.[404]

Specialized AI Models: Specialized AI models focus on specific tasks in clearly defined fields. They are trained with data from specific fields of application. One example is AI models for diagnostic imaging in healthcare. These must be distinguished from models derived from general generative models that are specialized in a certain way, e.g. for a specific language.

Inference: Inference is the process by which an AI model applies what it has learned and derives a result from a new input.

H100 Equivalent: H100 is a high-performance graphics processing unit (GPU) by Nvidia, which is primarily used for AI training. The performance of the H100, which has been on the market since 2022, is considered the benchmark for comparing the performance of different AI computing systems.[405]

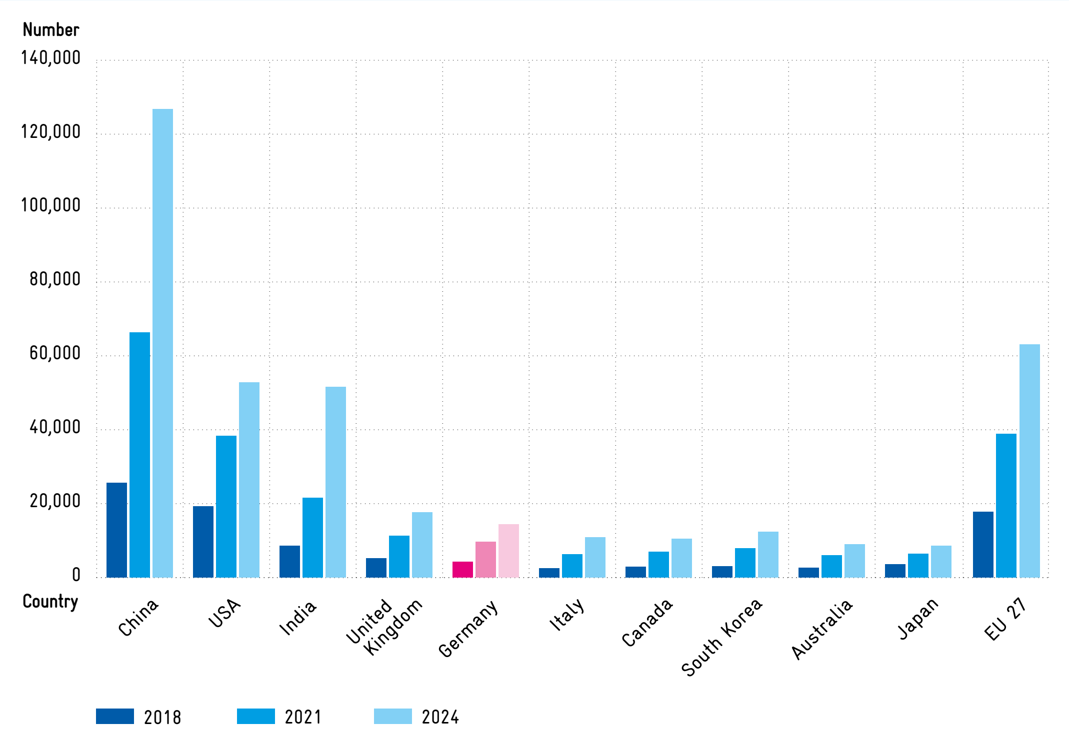
B 3-2 International Comparison of AI Research and Development

AI Publications: EU on Par with the USA

Scientific publications on AI have increased significantly worldwide since 2018 (cf. figure B 3-3).[406] In all three years under review, most publications came from China, followed by the EU, which had significantly more AI publications than the USA in 2024. The USA and India, in third and fourth place, are at a similar level to the EU. Germany ranks fifth behind the UK in terms of publications, making it the EU country with the most AI publications.[407]

Looking at the share of publications on generative AI in all AI publications (cf. figure B 3-4), it can be seen that in 2024, the USA had the highest share among the top 10 publishing countries with 16.9 percent, followed by Italy (15.3 percent) and Germany (13.4 percent). In China, the share was only 8.7 percent. In addition, the share

Fig. B 3-3 Number of publications by the top 10 countries in the field of AI in 2018, 2021 and 2024



Legend: In 2024, a total of 14,415 AI publications were published in Germany.

Due to differences in data sources, the figures presented here differ slightly from those in chapter A 2. The trends and the ranking of the countries are consistent.

Source: Elsevier SCOPUS. Calculations based on Weber et al. (2026).

of generative AI publications in China and India grew less strongly than in the other countries.[408] In absolute terms, however, China continues to clearly dominate the USA and the EU in terms of publications on generative AI. In absolute terms, Germany ranks behind India and the UK.

AI Patent Applications: USA and China Dominant

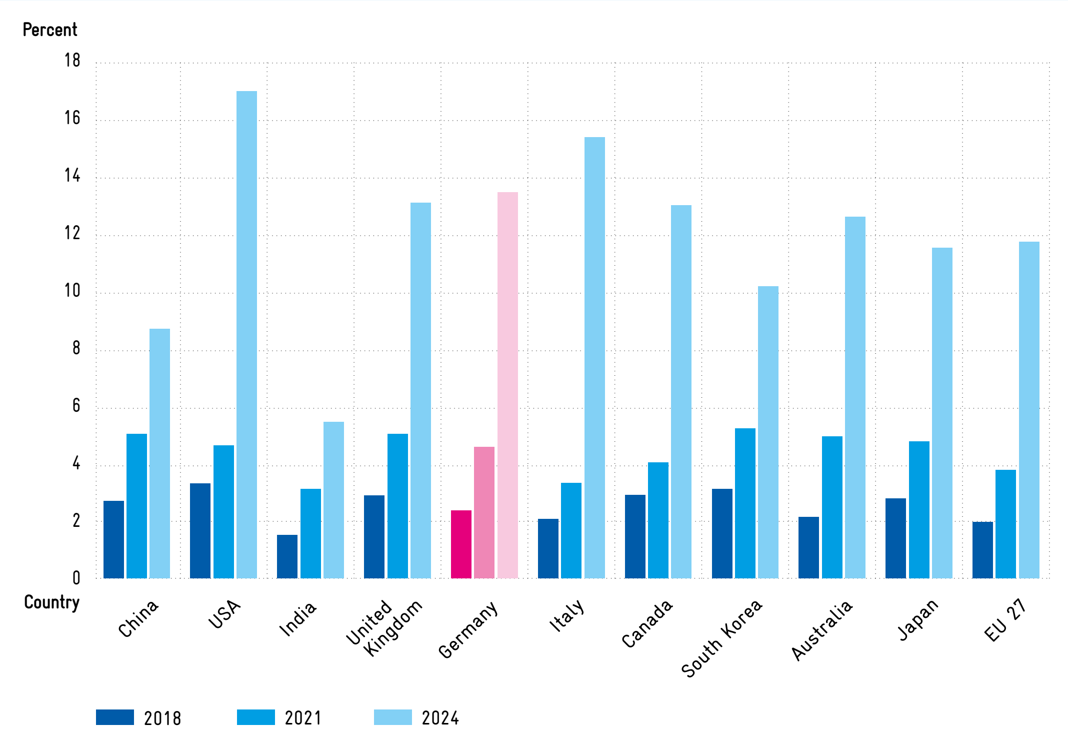
Figure B 3-5 shows transnational AI patent applications from the ten countries with the most patent applications. In China and the USA, the absolute number of transnational AI patent applications rose sharply after 2016. A more moderate increase can also be observed in the other top 10 countries. In the period from 2020 to 2022, the USA and China dominated ahead of the EU, Japan and South Korea. Germany lagged significantly behind Japan and South Korea but was the EU country with the most transnational AI patent applications. The UK, one of the biggest AI actors in Europe, ranks behind Germany.[409]

In Germany, most transnational AI patent applications come from large companies such as Siemens, Robert Bosch, Siemens Healthcare, Volkswagen and Bayer. For most of these companies, AI patents account for less than 10 percent of their total transnational patent applications; only Siemens Healthcare has a significantly higher share, at 25.7 percent.

USA Dominates Development of Notable AI Models

A market-orientated indicator of a country's innovative strength in the field of AI is the number of AI models developed. To this end, figure B 3-6 examines data from the non-profit research organization Epoch AI on notable[410] AI models.[411]

Fig. B 3-4 Share of publications on generative AI in total AI publications for selected countries in 2018, 2021 and 2024 in percent



Legend: In Germany, the share of publications on generative AI out of all AI publications stood at 13.4 percent in 2024.

The selection of countries is based on figure B 3-3.

Source: Elsevier SCOPUS. Calculations based on Weber et al. (2026).

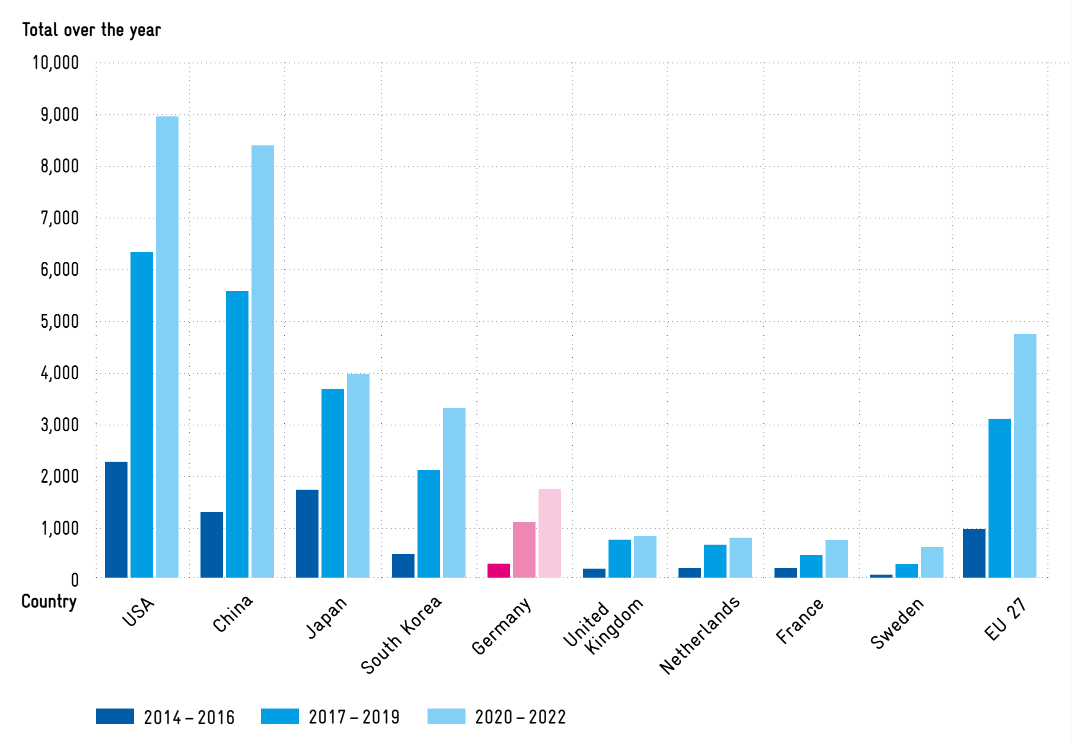
Figure B 3-6 shows an increase in the development of notable AI models in most countries, with the USA leading by a wide margin. China lags clearly behind the USA but is well ahead of the UK and the EU. Within the EU, 24 notable AI models were published between 2023 and 2025, eleven of which came from France and nine from Germany. Notable models from Germany originate primarily from research institutions that often collaborate with US partners, such as the Technical University of Berlin with Google, but not with organizations from other EU Member States. German companies that have published notable AI models include DeepL, deepset and Bosch.

Overall, these three analyses paint a familiar picture: Germany and the EU are strong in research, mediocre in patented inventions and comparatively weak in the development of innovative products.

Publicly funded AI Models in Europe

To support the competitiveness of European companies and research institutions, the development of AI models that pay particular attention to compliance with the EU Data Protection Directive is publicly funded in Europe. Examples include Teuken-7B from the OpenGPT-X research project of the Fraunhofer Institutes IAIS and IIS, and EuroLLM-9B from Utter, a joint project of various European universities. Apertus-70B from the Swiss AI Initiative is an open model whose parameters, data and training algorithms are publicly available; it was trained with the opt out consent of data owners and avoids storing training data in the model. With 7, 9 and 70 billion parameters respectively, these models are relatively small compared to the leading commercial models, which use more than 500 billion parameters.[412]

Fig. B 3-5 Number of transnational AI patent applications by the top 10 countries 2014–2022



Legend: Between 2020 and 2022, 1,709 transnational AI patent applications were filed from Germany.

Transnational patent applications are applications in patent families with at least one application to the World Intellectual Property Organization (WIPO) via the PCT procedure or one application to the European Patent Office. Due to differences in the scope of the data, the figures presented here differ slightly from those in chapter A 2. The trends and the ranking of the countries are consistent.

Source: PATSTAT. Own representation based on Weber et al. (2026).

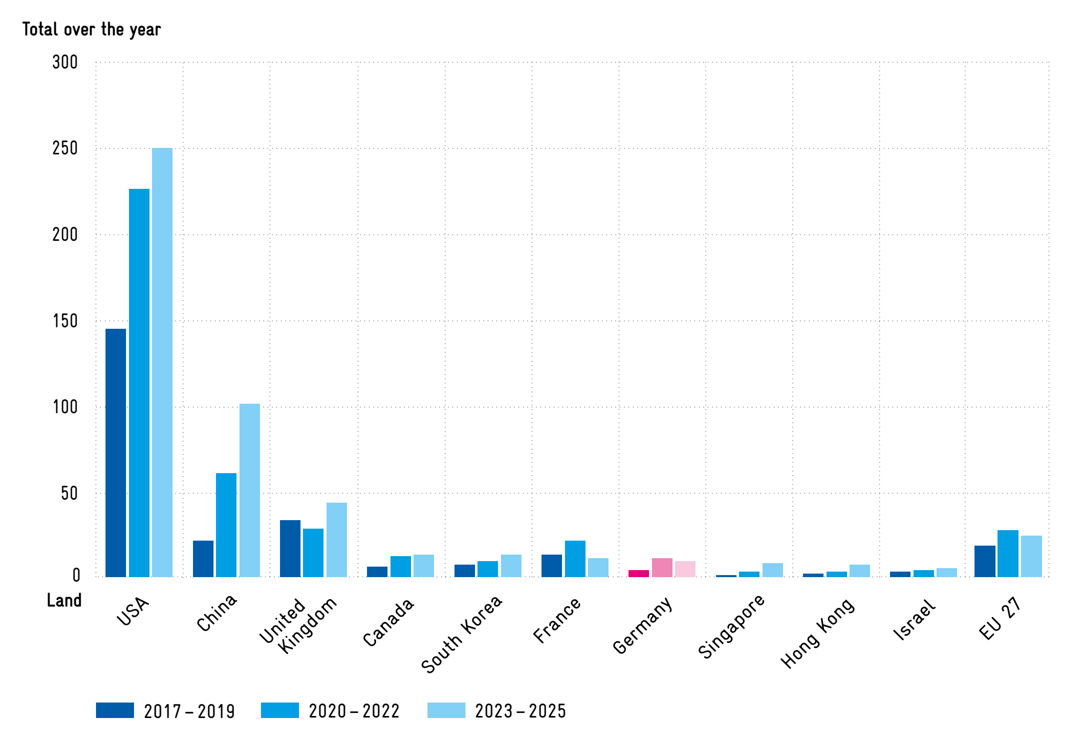
Some commercial AI models from European companies also received public funding. TFree-HAT-Pretrained-7B-Base from the German company Aleph Alpha was trained on English and German data and developed specifically for use in German-speaking countries. The multimodal open model Mistral Large 3 from the French company Mistral, released in early December 2025, exists in different variants with up to 675 billion parameters. Mistral models are often used as a basis for further developments. One such example is Minerva-7B, which has been completely retrained with Italian language data.[413] To develop future AI models in Europe, the Next Frontier AI initiative was launched, a project with which SPRIND aims to promote the development of next-generation AI companies.[414]

B 3-3 Use of AI in the Private and Public Sectors

Germany Ranks Mid-Range Globally in Terms of Number of AI Companies, but Leads the EU

A high number of specialized AI companies, i.e. companies that develop AI or whose products or services rely significantly on AI, is an indicator of innovative strength, economic dynamism and the ability to translate research results into marketable applications. With this in mind, a study commissioned by the Commission of Experts examined Germany's position in terms of the number of AI companies.[415] An evaluation of the Crunchbase database[416] from February 2025 shows that, although Germany is one of the leaders in terms of the number of AI companies within the EU, it only occupies a mid-range position in a global comparison and also

Fig. B 3-6 Number of notable AI models in the top 10 countries 2017–2025



Legend: Between 2023 and 2025, nine notable AI models were developed in Germany.

Source: Own calculations based on data from Epoch AI (2025).

lags behind smaller economies such as Canada and the UK (cf. figure B 3-7).

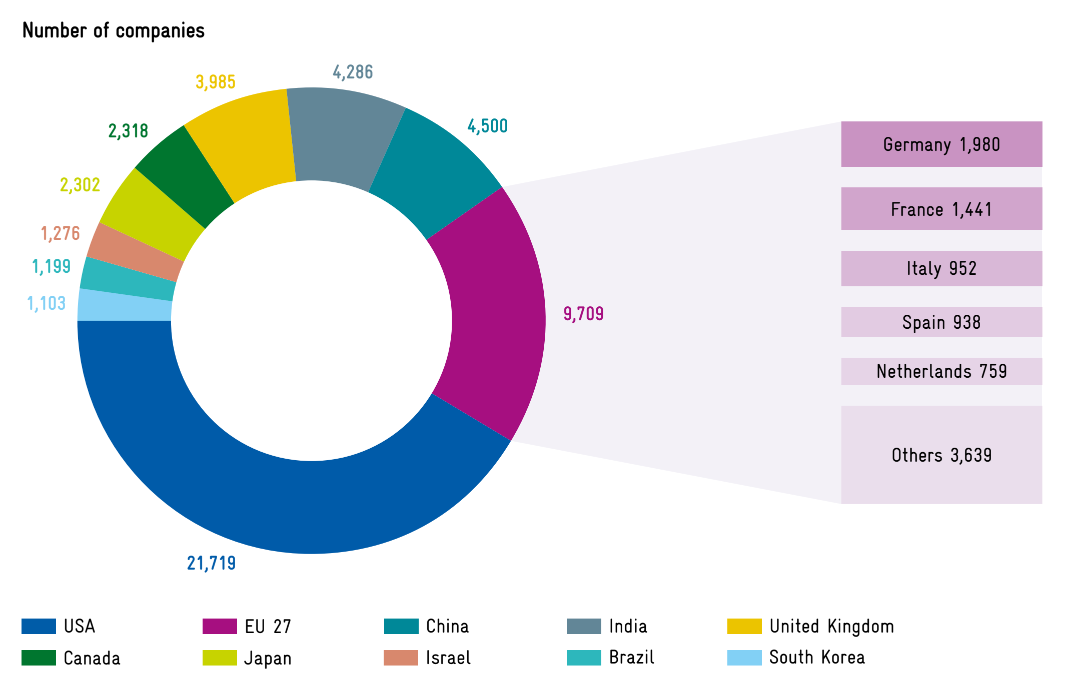
The USA is the clear global leader with 21,719 AI companies, ahead of the EU with 9,709 AI companies. Germany leads the EU ranking with 1,980 AI companies. In a global comparison, Germany ranks seventh, behind the USA, China (4,500 AI companies), India (4,286), the UK (3,985), Canada (2,318) and Japan (2,302), but ahead of France (1,441). Among the other EU Member States, Italy, Spain and the Netherlands in particular have a relatively high number of AI companies (between 759 and 952).

Start-ups, i.e. companies that are less than five years old, are a key driver of the development and spread of AI, especially when they quickly adopt new technological approaches, develop them at a rapid pace, scale them up quickly and transform them into marketable applications. At 31.3 percent, the share of start-ups among all AI companies in Germany is almost on a par with that in the USA (32.9 percent) and the UK (30.8 percent) and slightly above the EU average (27.7 percent). Italy (28.5 percent) and the Netherlands (27.9 percent) also show similar figures to the EU average. The results indicate that the German AI business landscape is characterized not only by a high number of AI companies, but also by active start-up activity in comparison to other European countries.[417]

Despite the positive picture painted by the analysis of Crunchbase data, (AI) start-ups in Germany and the EU face two major obstacles. Firstly, the fragmentation of the European Single Market represents a significant barrier to scaling up. Different national regulations and markets increase costs for start-ups and can hinder their growth. More integrated markets such as the USA or China have an advantage here. A ‘28th regime’, a project of the EU as part of its Start-up and Scale-up Strategy to reduce fragmentation in the EU Single Market, would remedy this situation (cf. chapter A 4).[418]

Another problem lies in the taxation of employee share ownership in Germany. So far, monetary benefits from asset participation have been taxable in

Fig. B 3-7 Number of AI companies worldwide listed on Crunchbase in February 2025



Legend: In February 2025, there were a total of 9,709 AI companies in the EU 27, 1,980 of which were based in Germany.

For the purposes of this analysis, AI companies are defined as those that, according to Crunchbase's own industry classification, belong to the 'Artificial Intelligence' industry group.

Source: Crunchbase. Calculations based on Weber et al. (2026).

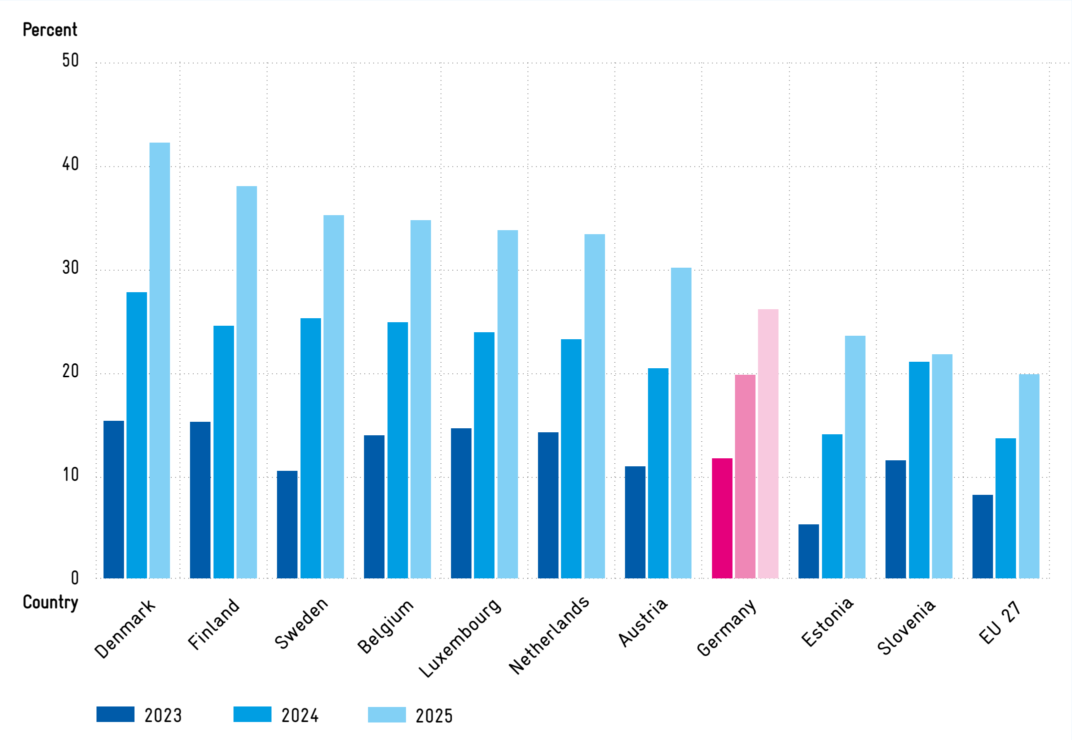
the event of a change of employer or after 15 years at the latest, even if no real inflow of liquidity has taken place (so-called dry income taxation). However, liquidity usually only flows when the company is sold or goes public.[419] This tax regulation, which is unattractive by international standards, makes it difficult for growth-orientated start-ups to attract highly qualified personnel.[420]

Diffusion of AI in the Economy Increasing Rapidly

The use of AI in the German and European economies has gained significant momentum in recent years (cf. figure B 3-8). Eurostat surveys show that in Germany, the share of companies with at least ten employees that use AI in general rose from 11.6 percent in 2023 to 26.0 percent in 2025.[421] In 2025, Germany ranked eighth in a European comparison. The Nordic countries Denmark (42.0 percent), Finland (37.8 percent) and Sweden (35.0 percent) have the highest shares here. The Benelux countries also have higher shares than Germany (Belgium 34.5 percent, Luxembourg 33.6 percent, Netherlands 33.2 percent).

According to the Eurostat survey, in 2025, the percentage of companies that generally use AI was highest in the group of large companies with 250 or more employees in all countries under review (Germany: 57.0 percent; EU: 55.0 percent), followed by medium-sized companies with 50 to 249 employees (Germany: 35.6 percent; EU: 30.4 percent) and small enterprises with 10 to 49 employees (Germany: 23.1 percent; EU: 17.0 percent). In addition, a representative survey conducted on behalf of the Commission of Experts also shows a rapid increase in the use of generative AI, particularly in the information economy and the manufacturing sector.[422] It should be noted, however, that these surveys do not record the type of use. There are major differences between the use of simple chat tools and that of agent-based generative AI. The development described therefore only allows conclusions to be drawn about a general trend in AI use, but not about changes in its quality.

Fig. B 3-8 Share of companies with more than ten employees using AI in the top 10 EU countries in 2023–2025 in percent



Legend: In Germany, 11.6 percent of companies with more than ten employees were using AI in 2023, whereas by 2025 this figure had risen to 26.0 percent.

Source: Own representation based on Eurostat (2025b).

AI in Public Administration: Many Questions Remain Unanswered

With its Modernization Agenda for the State and Administration, the German Federal Government has adopted an interdepartmental reform programme that aims to make the state "simpler, more digital and more successful" and focuses, among other things, on the use of AI.[423] Although 23 lever projects bundle measures that are considered particularly effective and can be implemented in the short term, key structural reforms have yet to be implemented. Furthermore, the Deutschland-Stack is intended to create a platform that offers a "secure, interoperable, European-compatible and sovereign" infrastructure for the digitalization of administration and integrates cloud services and AI applications.[424] This is a desirable goal, but the specific design remains unclear and should be determined as soon as possible. A uniform digital basic system would be a prerequisite for making AI services usable on a large scale and avoiding individual solutions. A modular approach that simplifies the replacement of individual software elements in the Stack can reduce dependencies through lock-in effects.[425] It must also be clarified to what extent the underlying infrastructure is actually designed to be "sovereign" and independent of non-European providers. Finally, clear impact indicators and concrete timetables would help to measure the innovative effects of AI in public administration and the intended strengthening of sovereignty.[426]

There are also impulses from the private sector for the digitalization of the public sector, for example to improve citizen services or optimize internal information flows. In autumn 2025, SAP and OpenAI announced the 'OpenAI for Germany' partnership, which aims to make large language models such as ChatGPT and other AI technologies available to the public sector via the Delos Cloud, in compliance with data protection, data sovereignty and compliance requirements in Germany.[427] Nevertheless, this solution cannot guarantee technological independence or data sovereignty, as the Delos Cloud is based on components of proprietary Microsoft Azure technology, which is subject to the US CLOUD Act.[428] Therefore, the data is not safe from access by US authorities, even though it is physically stored in data centres in the EU and is therefore subject to the GDPR. Added to this is the lack of transparency regarding the functioning and training data of OpenAI's models. In light of this, it seems appropriate to continuously evaluate the sovereignty and security architecture of such hybrid models.

B 3-4 Infrastructure and Compute Power

Germany and EU Lag Behind International Competitors in Compute Power

Both training and using large AI models require extensive compute power. The further development of AI systems and the provision of models for the mass market also place very high demands on compute power. Accordingly, the capacity expansions currently planned, particularly in the USA, go far beyond existing capacities.

Although investment in data centres in Germany reached an all-time high in 2025, it remains low by global standards.[429] Unlike in the USA, there are currently no very large data centres in Germany that are used exclusively for AI. Figure B 3-9 illustrates how much the USA dominates in terms of compute power, with approximately six times the capacity of China.[430] The capacities of European countries, including Germany with around 26,000 publicly recorded H100 equivalents, are lower by several orders of magnitude. In addition, compared to the USA, where most of the AI compute power is built, financed and used by the private sector, most of the AI compute power in Germany is publicly operated, meaning that, due to EU state aid law, private companies can only use it in the pre-competitive field.[431] Noteworthy private sector projects include the announced data centre of Deutsche Telekom with an estimated capacity of around 20,000 H100 equivalents and that of Schwarz Digits with a capacity of up to 100,000 GPUs.[432]

Bottlenecks in Expansion of AI Compute Power in Germany

The construction of new data centres in Germany and the EU is currently being hampered by bottlenecks in the power grid, high electricity costs and administrative barriers. Between 2010 and 2024, the electricity consumption of German data centres rose by around 90 percent despite significant efficiency gains;[433] the increasing use of AI applications is likely to cause energy demand to rise significantly further.[435] In Germany, the lack of suitable grid and connection capacities is a bottleneck for the expansion of AI and data centre infrastructure,[436] especially in conurbations such as the Rhine-Main area and Berlin, where land and grid capacities are particularly limited.[437] In addition, the high cost of electricity compared to other countries is a competitive disadvantage, as it can account for around half of the operating costs of data centres.[438] France and the Nordic countries of Denmark, Sweden, Norway and Finland have a more cost-effective, climate-friendly energy supply and network capacities[439] that additionally favour the rapid development of high-performance compute power.

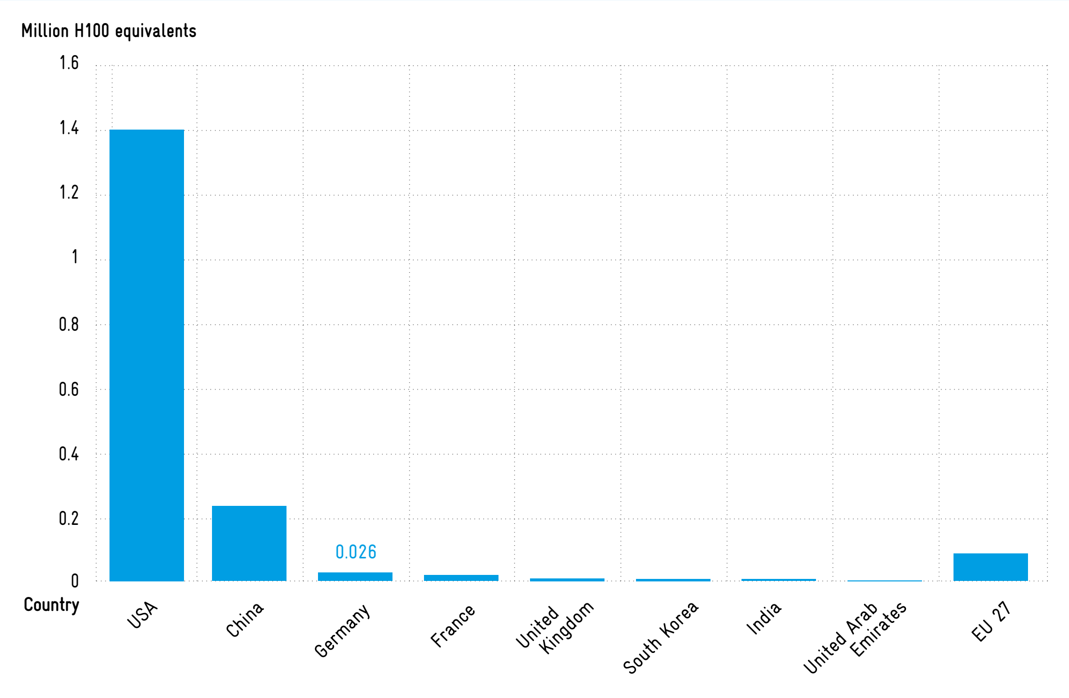
In addition to energy bottlenecks, lengthy approval and planning procedures are hampering the expansion of data centre infrastructure.[440] The establishment of new data centres is being delayed by complex environmental, construction and grid connection procedures.[441] International comparisons reveal locational disadvantages in this area: France and Norway, for example, have partially accelerated and better coordinated approval and planning procedures.[442] On average, it takes up to seven years in Germany to connect a data centre to the grid – significantly longer than in other countries.[443]

AI Gigafactories Planned to Support Expansion of AI Capacity in the EU

The European Commission's AI Continent Action Plan covers five fields: infrastructure, data access, algorithms, AI skills and simplification of legislation.[444] The core element of the infrastructure field is the development of AI compute power in so-called AI Factories and AI Gigafactories.

AI Factories are AI ecosystems centred around European supercomputers that are part of the European High Performance Computing Joint Undertaking (EuroHPC JU).[445] They pool material and human resources and are primarily intended to serve the development and training of AI models, in particular by start-ups, small and medium-sized enterprises (SMEs) and academia.[446] In December 2025, nine of these AI Factories were in operation,

Fig. B 3-9 Publicly documented compute power in large data centres for selected countries in 2025 in millions of H100 equivalents



Legend: In Germany, around 26,000 H100 equivalents were recorded in publicly documented large data centres for the year 2025.

These figures should be interpreted in relative terms, not as exact inventory figures. The underlying data from Epoch AI covers only 10 to 20 percent of all global large-scale data centres. As the data is based on publicly available sources, it covers public and private computing infrastructure to varying degrees.[434]

Source: Epoch AI. Own calculations based on data by Pilz et al. (2025).

including the JAIF (Jupiter AI Factory, Jülich) with approximately 24,000 H100 equivalents[447] and the IT4LIA (Italy for Artificial Intelligence, Bologna) facility, which currently has around 6,000 H100 equivalents.[448] Further AI Factories are in the planning stage. When setting up the planned AI Factories, it makes sense to promote innovative solutions from young European companies. One example is the photonic coprocessors developed by the German company Q.ANT, which are already being used in public data centres.[449]

AI Gigafactories are five large-scale facilities planned by the European Commission, each with approximately 100,000 H100 equivalents,[450] where complex AI models are to be developed, trained and deployed. After 76 consortia expressed interest in operating one of these Gigafactories in mid-2025,[451] the invitation to tender is expected early this year. Given an estimated investment volume of €3 to €5 billion per AI gigafactory, the European Commission plans to operate them as public-private partnerships and cover up to 35 percent of the initial investment.[452] A European Union regulation from December 2025 stipulates that a maximum of 17 percent of investment expenditure should be financed from EU funds. The participating Member States are to contribute at least the same amount.[453] In return, a corresponding portion of the compute power is to be made freely available for public applications.

However, there are doubts about the planned implementation of the AI Gigafactories.[454] The Commission of Experts shares this scepticism to a certain extent and notes that there is still a need for clarification, particularly with regard to the following points:

— Costs: The EU plans to fund only the initial investment in AI Gigafactories with up to 35 percent from public funds. In addition, this share is to be offset by AI computing power.[455]

However, the ongoing operating costs of AI data centres are considerable, particularly due to their high energy consumption. This is particularly significant in locations with high electricity costs, such as Germany. In addition, regular replacement investments are required, as GPUs can only be used for three to five years, depending on their utilization, and become technically obsolete even faster.

It is true that companies and consortia that submit detailed proposals for the operation of AI Gigafactories will have analyzed their economic viability and developed viable business models.[456] However, given the high level of uncertainty, the risk of misinvestments or politically motivated decisions remains. In addition, the EU could find itself under pressure if, after a few years, the Gigafactories can only continue to operate with additional funding.

— Location: Within the EU, potential locations for the planned AI Gigafactories differ significantly in terms of energy costs.[457] The extent to which factors such as local demand, operator expertise and the availability of complementary services can justify a location with higher electricity costs remains an open question. The identification of a suitable location in terms of electricity prices and sufficient connection capacities is crucial for a potential AI Gigafactory in Germany.

— Use: The planned AI Gigafactories are to be used for both the development and training of AI models and for inference.[458] This seems sensible, as the planned capacity is unlikely to be utilized to full capacity in the long term through development and training alone. The large AI compute power in the USA is used to a large extent by companies with extensive AI activities such as OpenAI, Anthropic, Google and Meta, which currently have no equivalent in Europe. Users of AI Gigafactories will therefore often be companies with little AI experience. For them, Gigafactories must provide not only AI compute power, but also supporting services such as complementary software and consulting. The Gigafactories should also be closely linked to existing research, expertise and start-up ecosystems in order to enable the development of know-how and effective use of infrastructure.

The specialization of AI Gigafactories according to function (development and training versus use) or field of application (e.g. industry or medicine) could help to achieve a critical mass of expertise. For inference, it should also be noted that AI Gigafactories will face global competition and therefore require efficiency-orientated business models.

B 3-5 Investment in AI

AI Expenditure in Germany Strongly Focused on Research

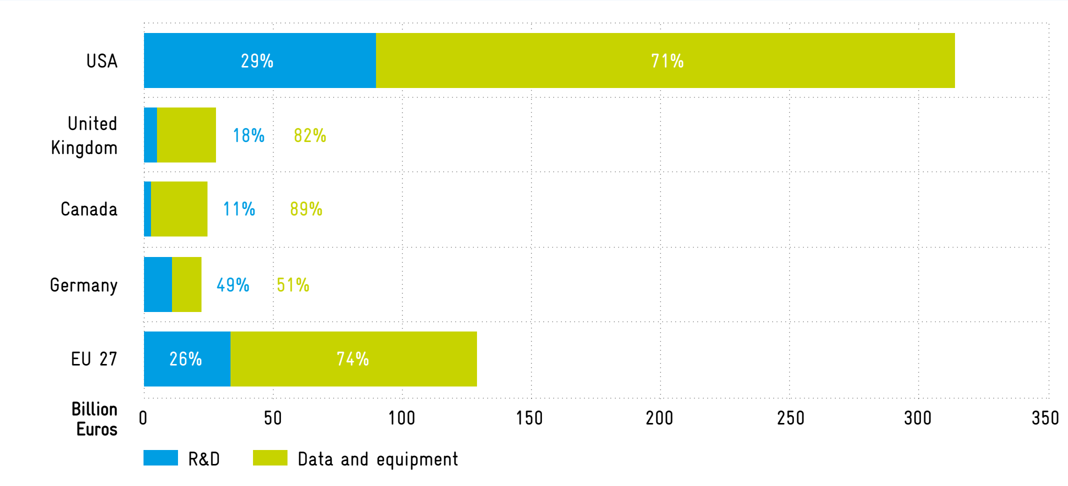
Precisely measuring the volume of private and public investment in AI is methodologically challenging.[459] A recent study estimated this expenditure for 2023 for EU Member States and, where data availability permitted, for selected non-European countries.[460] AI-related expenditure on research and development (R&D)[461] and expenditure on data and equipment[462] (cf. figure B 3-10) can be compared internationally. This shows that the EU, with a total of around €130 billion (0.75 percent of GDP), lags significantly behind the USA, with €310 billion (1.22 percent of GDP). With estimated expenditure of around €20 billion, Germany makes a significant contribution to EU-wide AI expenditure in absolute terms, but at 0.53 percent of GDP, it is below the EU average and far behind the USA.[463] In addition, investment activity in 2025 has exacerbated this investment gap with the USA, with investment rising significantly in the USA and more slowly in Germany and the rest of the EU.

In Germany, R&D expenditure accounts for a relatively high proportion of total expenditure at 49 percent (cf. figure B 3-10). In an international comparison of AI-related R&D expenditure, the EU (€33 billion, 0.19 percent of GDP) lags far behind the USA (€90 billion, 0.35 percent of GDP).[464]

European Venture Capital Investment Has Room for Improvement

Another indicator of innovation-driving expenditure on AI that allows for international comparison is the volume of private venture capital investment. A look at private venture capital flows between the five regions that account for the most AI-related venture capital worldwide reveals significant differences. In 2025, EU investors invested significantly less venture capital in AI (US$13.1 billion)

Fig. B 3-10 Estimated AI-related expenditure in selected countries in 2023
in billions of euros



Legend: In Germany, AI-related expenditure totalled €22.1 billion in 2023. Of this, 49 percent was spent on research and development.

Source: Own representation based on OECD estimates in Fonteneau et al. (2025).

than investors in the USA (US$119.8 billion), the UK (US$19.4 billion) and China (US$16.3 billion) (cf. figure B 3-11).[465] Possible reasons for this include the fragmentation of the European capital market, restrictive investment regulations for insurance companies, people's saving behaviour and a pension system that is largely based on pay-as-you-go/earn financing. A European Capital Markets Union, also known as a Savings and Investment Union, could remedy this situation and significantly improve opportunities for start-ups in the EU.[466]

In terms of total venture capital investment in AI, EU-based companies rank second with approximately US$13.8 billion, but are roughly on a par with the UK (US$12.1 billion), which attracts a comparable volume of investment despite its significantly smaller market. At 53 percent, the largest share of venture capital investments received by EU-based companies comes from the EU itself, followed by the USA (31 percent) and the UK (15 percent). The USA is the economic area that attracts the most AI-related venture capital.

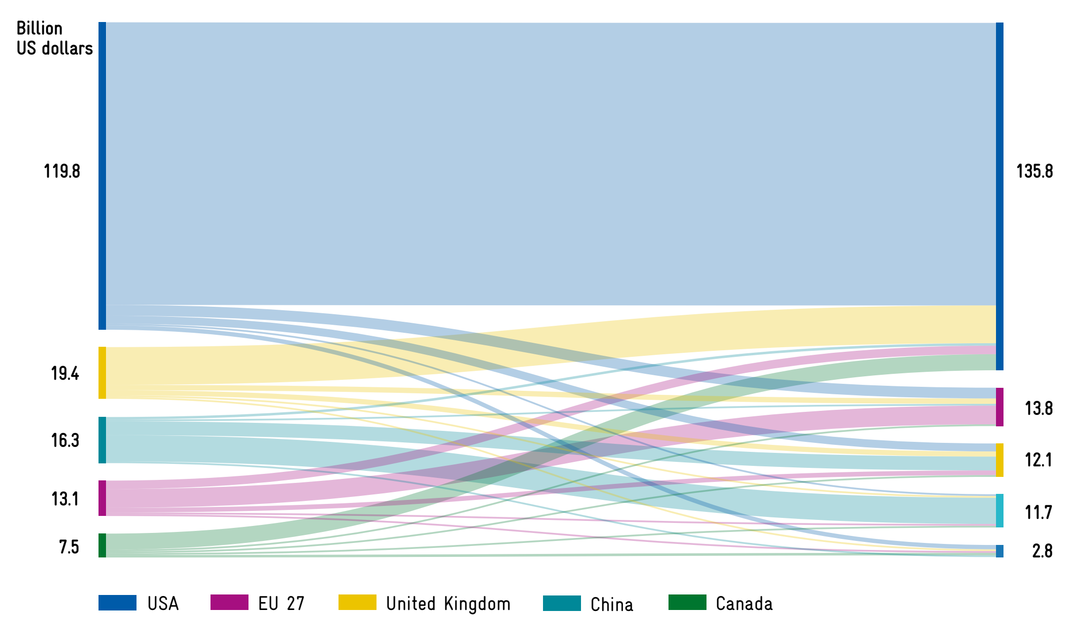
Fragmented Public Funding

A look at public investment in AI reveals a wide variety of support programmes at European, national and regional level.[467] Funding for AI research within the EU's Research Framework Programme has increased significantly in recent years, both in terms of the number of projects funded and the volume of funding.[468] In addition, there are targeted digital funding lines at EU level that are more application-orientated, such as the Digital Europe Programme and the European Investment Bank's TechEU programme. Germany's national AI Strategy had a budget of around €5 billion available between 2018 and 2025, of which only €2.8 billion had been used for specific projects by the end of 2024.[469] Germany's High-Tech Agenda also provides for funding of around €1.5 billion for AI. No reliable evaluations are available to date that allow causal statements to be made about the effectiveness of the funding used.[470]

B 3-6 Data Availability

The development of powerful AI models requires extensive, high-quality data. Both broad-based and specialized data sets are important: general data (e.g. general text, speech and image data) provides a basis for foundation models, while domain-specific applications require specialized data (e.g. industry-specific production, sensor and personal data). Making suitable training data available is a challenge. Many experts see the lack of data availability, both for general and specialized models, as the biggest obstacle to AI in Europe, even more so than limited compute power.[471]

Fig. B 3-11 AI-related venture capital flows between regions in 2025
in billions of US dollars



Legend: By 2025, European investors had invested approximately US$13.1 billion in AI venture capital. During the same period, the EU 27 received approximately US$13.8 billion in AI-related venture capital investments. Just under 50 percent of this came from the EU.

The selected regions are the five country groups from which the most AI-related venture capital originates. The analysis covers only AI-related venture capital flows between the five country groups under consideration. Investments from or into countries outside these groups are not taken into account. Last revised 5 January 2026.

Source: Own representation based on data from Prequin and processed by OECD.AI (2026).

Development of AI Foundation Models in the EU Complicated by GDPR

US-American AI companies typically use publicly available internet data to train large AI models. Legal uncertainties and any resulting legal proceedings[472] seem to be accepted in the interests of rapid market readiness. In Europe, on the other hand, the GDPR limits the use of internet data beyond copyright protection. Added to this is uncertainty about the interpretation of the laws; for example, it is considered unclear whether and to what extent AI training with data from the Common Crawl Foundation[473] is permitted. These restrictions hinder innovation. For example, there are foundation models from German research where legal concerns regarding the GDPR have led to delays of several months and the models ultimately being published only under a research licence, thus excluding commercial use.

According to legal experts, a strict interpretation of the GDPR prohibits the use of personal data for the training or operation of AI systems.[474] With its prohibition principle and reservation of permission, the GDPR thus worsens the framework conditions for the development of data-intensive AI models in Europe. The European Commission's Digital Omnibus, presented in November 2025, proposes exceptions which, according to legal expert Härting, do not solve the problem for companies: "The confusing web of exceptions and exceptions to these exceptions in Art. 9(5) GDPR will lead to the recommendation that the 'safest course' is to completely refrain from using specially protected personal data when training and operating AI applications."[475]

It seems downright absurd that, at the same time, US providers are successfully marketing their AI models in Europe, even though their model training practices do not comply with the requirements of the GDPR.

Scepticism About Data Exchange Slows Down Development of Specialized Models

The data required for the development of specialized AI models in industry is mostly proprietary and in the hands of various companies. However, effective AI training often requires combining data from different companies, which is potentially problematic for the protection of trade secrets as well as for data protection. One solution lies in the use of privacy-enhancing technologies (cf. box B 3-12). Another approach is the use of trusted data intermediaries (TDIs). These include data trustees, data rooms such as Catena-X, a data ecosystem for the automotive industry, and data marketplaces.

Various initiatives already exist to simplify data sharing. For example, acatech's Mission AI aims to connect data rooms across industries and countries, thereby strengthening the database, standards and quality checks for AI applications in Germany.[476] As part of this, for example, they have developed a data set search engine[477] and an AI-based compliance testing system for secure data exchange.[478] Another example comes from the mechanical engineering sector in the form of the Siemens Data Alliance.[479]

According to a Bitkom survey conducted in April 2025, 9 percent of the companies surveyed used data rooms, with a further 40 percent considering doing so.[480] As part of the Important Project of Common European Interest (IPCEI[481]) Industrial AI, there are plans to improve the availability of data from industry – among other things, linking up with the Manufacturing-X data room initiative.[482]

As part of its digital strategy, the EU has already drafted several laws aimed at strengthening the EU's data economy. For example, the Data Governance Act aims to strengthen data exchange by creating transparency and security for actors.[483] The Data Act, in turn, aims to ensure fair access to and fair use of data (in particular data generated through the use of networked products and connected services).[484] In addition, there are sector-specific data laws such as the European Health Data Space Regulation. However, various contradictions exist between all these laws,[485] meaning that the legislation currently fails to achieve its purpose of simplifying data sharing. The European Commission has now recognized this problem. Harmonizing and simplifying data legislation[486] is one of the three pillars of the European Data Union Strategy published in November 2025, which in turn has the explicit goal of making data usable for AI.[487]

B 3-7 Regulatory Framework and Funding Opportunities

Work Underway to Simplify Implementation of the AI Act and Other Digital Laws

The AI Act, the centrepiece of European AI legislation, has been in the process of implementation since the beginning of this year. Since August 2025,

Box B 3-12 Examples of Privacy Enhancing Technologies for Secure Sharing of AI Model Training Data

Multi-Party Computation: A cryptographic method in which multiple parties jointly perform calculations on their data without having to disclose their raw data to each other.

Federated Learning: A distributed learning method in which the AI model is brought to the data and not vice versa. The data remains decentralized (e.g. on devices, in clinics, companies). Only model parameters are shared, not the original data.

Synthetic Data: Artificially generated data that corresponds to real data in its statistical properties but does not contain any real personal information. It is used to train or test machine learning models when real data is inaccessible, too sensitive, incomplete or expensive.

Trusted Execution Environments: Hardware-based security areas in processors that protect data during processing, shielded from operating systems, cloud providers, and administrators.

its provisions have applied to general-purpose AI, and guidelines and a code of practice have been published to ensure compliance.[488] The majority of the AI Act's provisions will come into force in August 2026, when enforcement will also begin. However, the harmonized technical standards developed by the European standardization committees CEN and CENELEC on behalf of the European Commission to support the implementation of the AI Act have not yet been finalized. As the development of standards is delayed, companies fear that they will not have enough time to implement them before the AI Act comes into force. At the same time, it is becoming apparent that parts of the AI Act are extremely costly to implement – for example, transparency requirements, double regulation and coordination with existing digital laws.[489] There are also concerns that the standardization processes provided for in the AI Act do not take sufficient account of SMEs and start-ups.[490]

To simplify digital legislation, the European Commission presented the Digital Omnibus package on 19 November 2025.[491] The first sub-package amends key existing digital regulations such as the GDPR, the ePrivacy Directive and the Data Act. The second sub-package contains amendments and simplifications to the AI Act. Among other things, it stipulates that the obligations for high-risk AI systems will no longer apply automatically from 2 August 2026, but will be linked to the availability of technical standards.[492] In addition, some SME exemptions have been extended to small mid-caps, and an AI regulatory sandbox at European level is being envisaged, parallel to the national regulatory sandboxes provided for in the AI Act.[493]

Since the publication of the Omnibus proposal, there has been debate as to whether it actually only includes simplifications or also substantial deregulations that could weaken fundamental and civil rights. Critical voices are coming in particular from parts of the European Parliament and civil society.[494] In view of the controversial debate, it remains to be seen whether the Digital Omnibus package will be adopted before the general application date of the AI Act on 2 August 2026. If no agreement is reached, the previous deadlines of the AI Act will initially apply, which makes planning difficult for companies.[495] In this context, it is questionable whether a rapid agreement can be reached that can create legal certainty in the short term for companies and developers of high-risk AI systems. However, further delays due to ongoing regulatory uncertainty carry the risk of slowing down innovation and encouraging poor decisions.[496]

In August 2025, the key European governance requirements of the AI Act came into effect, including the establishment of the AI Board, the Scientific Panel and the Advisory Forum.[497] At the operational level of the EU, the AI Act Service Desk and the Single Information Platform were launched on 8 October 2025.[498] They are intended to support the consistent implementation of the regulation in the Member States by pooling the information resources of the national supervisory authorities and supplementing them with practical tools, such as a compliance checker.[499]

With regard to the competent German authorities, the Federal Ministry for Digitalization and State Modernization (BMDS) presented a draft bill in September 2025 that provides for a hybrid governance structure.[500] The Federal Network Agency (Bundesnetzagentur, BNetzA) is to play a pivotal role in this, serving as the market surveillance and notifying authority, the central point of contact for the EU Commission and the operator of a complaints office. In addition, the Coordination and Competence Centre for AI Regulation (KoKIVO) within the BNetzA is to provide support in interpreting horizontal legal issues. Specialist authorities such as the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) will remain responsible for sector-specific applications. It is currently unclear how cross-authority coordination will be organized, how much autonomy the specialist authorities will have in decision-making, and whether sufficient staff capacity will actually be built up at the authorities involved (especially at the BNetzA).[501]

Public Support Services for Introduction of AI in Companies

One important project supporting the use of AI in businesses is the IPCEI on industrial AI, which was launched in November 2025.[502] Coordinated by the Federal Ministry for Economic Affairs and Energy (BMWE), 13 EU countries are working together to develop new sovereign foundation models and other specialized models for industry. The IPCEI makes it easier to promote cross-border projects between different Member States. Germany is providing €1 billion in funding for this initiative. Companies have been able to apply with project proposals since early December 2025.

The use of AI in Mittelstand companies is supported by programmes such as the AI Competence Centres[503] funded by the Federal Ministry of Research, Technology and Space (BMFTR) and the Mittelstand-Digital Innovation Hubs funded by the BMWE, as well as by state initiatives such as regional AI labs. All these programmes are designed to boost knowledge transfer, lower financial barriers, build skills and support Mittelstand companies in the practical introduction of AI technologies.[504]

The European Digital Innovation Hubs (EDIH), co-financed by the European Commission and EU Member States, also aim to achieve this.[505] According to a study by the Joint Research Centre (JRC), they fulfil their task of providing companies with knowledge about the possible applications of AI.[506] However, there is still room for improvement in terms of links to regulatory authorities, providers of AI regulatory sandbox and standardization authorities. At present, EDIHs can only serve as a first point of contact for regulatory support to a limited extent and should be developed iteratively to fulfil this task.

B 3-8 Digital Sovereignty

Europe's digital sovereignty is a declared goal of the Federal Government[507] and an essential component of the more general concept of technological sovereignty, which can be defined as follows:

"A national economy has technological sovereignty if it can itself provide and further develop a technology it deems critical for its welfare, competitiveness and ability to act, and if it can participate in its standardization and is able to apply and to source this technology from other economic areas without one-sided structural dependency."[508]

In this sense, digital sovereignty refers to digital technologies and digital security.[509] As a key enabling technology, AI is one of the technologies that is relevant in terms of digital sovereignty.

Digital sovereignty is only possible under the following conditions:

— Technology: Avoiding unilateral dependencies on non-European technology providers, including in the fields of cloud infrastructures, operating systems, artificial intelligence and semiconductors.

— Regulation: The ability to set standards instead of adopting the rules of other countries.

Digital sovereignty, in turn, is a prerequisite for data control, security and the preservation of social values:

— Data: Individual control over the data of European citizens and companies.

— Security: Cybersecurity, protection of critical infrastructure and digital defence capability.

— Society: Preservation of European culture and values such as data protection, freedom of expression and democratic control.

The prerequisites for digital sovereignty often go hand in hand with strong digital value creation.

Europe's Low Level of Digital Sovereignty Problematic

Europe's digital sovereignty is severely limited. Technologies from non-European companies such as AWS, Microsoft, Nvidia and OpenAI dominate the digital field, while Germany and Europe lag behind in AI patents and models (cf. B 3-2). With regard to data control, the GDPR and the possibility of data localization within the EU create a certain degree of security, but this is incomplete. Cybersecurity in Germany and the EU is limited because central digital infrastructures and security-relevant technologies are not sufficiently under their own control. Economically, the EU is heavily dependent on US providers. In 2024, the EU's deficit with the USA in trade in services, most of which were digital, amounted to €148 billion.[510] Added to this is a deficit of €52 billion for the eurozone alone in the transfer of primary income to the USA,[511] with cross-border payments from subsidiaries of large tech companies to their parent companies playing an important role. Finally, European culture and values are threatened by non-European content providers and by AI systems trained with non-European data.

The consequences of this limited digital sovereignty of the EU are highly problematic: access to technologies from non-European providers is not permanently guaranteed, which makes Europe vulnerable to blackmail. For example, under pressure from the US government, Microsoft blocked the email access of the chief prosecutor of the International Criminal Court in May 2025.[512] In addition, US President Trump announced in November 2025 that he wanted to prevent the sale of Nvidia's latest GPU abroad.[513] Even in data centres within the EU that are operated by European companies but use technical solutions from US providers, the security of European data is not fully guaranteed due to the provisions of the US CLOUD Act.[514] With regard to regulation, the EU appears to be sovereign, but in reality it can be put under pressure due to its dependence on non-European technologies.[515] The economic imbalance between the EU and the USA in digital services exacerbates this problem and also reflects the EU's lack of competitiveness and value creation in the digital field. The problems of limited cybersecurity and, ultimately, a threat to European culture and values are obvious. A serious example of the latter is the denial of the Holocaust by Grok, the AI model developed by the company xAI.[516]

The EU's sovereignty, especially in AI infrastructure, varies significantly depending on the component. The situation is particularly critical for GPUs, whose market is dominated by a single US company, NVIDIA.[517] Although data centre operators have the option of using products from other suppliers, these also originate from the USA. This unilateral dependence on US companies is increasingly being exploited politically, even though European companies are technological market leaders in upstream key enabling technologies for semiconductor manufacturing, which could be used as leverage in global value chains.[518] Examples of this are ASML's EUV lithography and the optics and laser systems from Zeiss and TRUMPF that are essential for this. The situation with high-performance AI models is similarly problematic to that with AI chips. These models originate almost exclusively from US and Chinese companies, and their use results in a worrying lock-in effect.

The situation is somewhat better for other hardware, especially data storage devices. Although the most important providers are mainly based in the USA and South Korea, there is a competitive market that reduces dependencies. Where supplementary software is concerned, e.g. for data management, partially sovereign European solutions do exist. European solutions are generally available for data centre operators and data centre locations. To improve the digital sovereignty of the EU and Germany in the field of AI, the EU is focusing on the planned AI Gigafactories (B 3-4) and AI models developed with public support (B 3-2).

Consistent Efforts Needed to Build Digital Sovereignty

Digital sovereignty cannot be achieved abruptly but is the result of a long-term and consistent plan. From the Commission of Experts' point of view, the issue of digital sovereignty is essential for Germany and the EU, as technological dependencies have a direct impact on political, economic and social scope for action.

When it comes to AI, the EU faces a strategic balancing act: on the one hand, it is important to avoid technological dependencies, but on the other hand, the use of AI must not be delayed by excessive caution and a lack of European solutions. A balanced approach is therefore necessary – temporarily accepting dependencies where they are unavoidable, while at the same time working systematically to reduce and ultimately overcome them in the long term.

Closer cooperation within the EU has a key role to play in this regard. A prime example of this is the EU AI Champions Initiative, which is being driven forward by business and politics in France and Germany and under whose umbrella 18 new AI-related corporate partnerships were formed in November 2025.[519] However, sceptical questions have been raised about how the intended cooperation between companies, some of which are direct competitors, can be structured in order to develop strategically important technologies.[520]

The issue of digital sovereignty is particularly evident in the field of internal security. In Germany, the use of Gotham analysis software from the US company Palantir is the subject of heated debate. Gotham accesses large amounts of data, e.g. from police authorities, analyzes and links it, and generates comprehensive situation reports (although the software's AI functions have been restricted or removed in Germany due to legal requirements).[521] While the Federal Ministry of the Interior considers automated data analysis and the use of AI to be indispensable,[522] data protectionists have expressed concerns about Palantir's proximity to US-American intelligence agencies and military institutions.[523] In addition, the far-reaching dependence on a US company in an application that is central to security policy must be viewed critically.

B 3-9 Recommendations for Action

Artificial intelligence has rapidly developed into a key enabling technology of the 21st century. It is increasingly shaping economic performance, international competitiveness and social innovation. The analysis clearly shows that although Germany and Europe have strong research landscapes and have achieved initial successes in the application of AI, they lag behind in the development of AI models and the implementation of AI in value creation when compared internationally, especially with the USA and China.

There is a lack of powerful infrastructure that reliably supports research, development and widespread application of AI, thereby strengthening Europe's digital sovereignty. The dominant role of non-European technology providers and the economic imbalance in digital services create structural dependencies that, in view of growing geopolitical tensions, call into question reliable access to key technologies and thus further jeopardize Europe's digital sovereignty. Furthermore, energy prices for AI development and operation are increasingly becoming a location factor.

There are also gaps in data availability, particularly in the industrial field and the public sector. In addition, regulatory requirements restrict the use of data for AI training in the EU more than in non-European countries. Last but not least, the EU lags behind the world's leading countries in terms of venture capital for AI. This, like the fragmentation of the Single Market, is an obstacle, especially for start-ups.

Given this situation, the Commission of Experts recommends a European approach and coordinated AI policy, expanding AI infrastructure, supporting top-notch research, having rules that are friendly to innovation, specifically backing the broad economic use of AI in Germany and the EU, and supporting start-ups in the AI field and their growth.

Taking a European Approach to AI Policy

— To catch up with the USA and China and avoid further technological dependencies, AI must be considered a key enabling technology at European level. Cooperation between Germany and France has a special role to play in this regard, acting as a kind of European driving force.

  Coordinated pooling of resources and expertise at EU level is crucial for an agile and effective European AI policy. On the one hand, funding structures must be harmonized horizontally and vertically, and the conditions improved through infrastructure investments. On the other hand, it also means distancing oneself from national interests and allowing intra-European competition so that European markets can be established.

  With this in mind, Germany should act as a strong voice at EU level for cooperation in building infrastructure, creating legal certainty and promoting AI in research and industry, and, as the largest EU country, take the lead in shaping the AI market in a uniform European and innovation-friendly manner.

— To strengthen the EU's position in global competition, Germany should strive for closer cooperation at EU level with strong democratic partners such as the UK, Canada, Japan and South Korea in order to jointly establish a third AI hub alongside the USA and China.

— To increase the total investment volume in AI in the EU, private investment is urgently needed in addition to public funding. To this end, it is essential to create more business-friendly conditions, for example through a 28th regime.

— Digital regulations at EU level (Data Act, AI Act, GDPR, etc.) should be harmonized in their national interpretations, contradictions should be eliminated, and compliance requirements should be simplified, especially for SMEs and start-ups. With this in mind, the Federal Government should advocate for further simplification of the AI Act when designing the Digital Omnibus. Regulatory uncertainty should be reduced as quickly as possible, as it slows down innovation and leads to wrong decisions.

Expanding European AI Infrastructure

— Germany and the EU should pursue dynamic expansion targets when building data centres at the European level. To keep up internationally, the EU should set itself the goal of providing 10 to 15 percent of global compute power within the next five years. To this end, the private sector must be empowered to drive forward expansion rapidly. In Germany, this requires simplified and accelerated approval procedures for data centres and an expansion of the energy networks.

— The framework conditions for Gigafactories must be designed in such a way that they are commercially competitive and do not require long-term subsidies. This requires a time limit on the compute power to be made freely available for public applications. The choice of location should be based on economic criteria in order to avoid politically motivated decisions favouring economically unsustainable solutions. The Commission of Experts recommends a phased approach to the construction of the planned AI Gigafactories in the EU so that learning effects can be exploited, flexibility is maintained and technological developments can be responded to.

— If data centres are built with public funds (as is the case with the EU's AI factories and gigafactories), a monitoring system should be set up to record utilization and types of use in particular and to ensure that capacities are used in a targeted manner.

— Given the short lifespan of GPUs, a shorter depreciation period seems sensible. To make the use of data centres attractive to businesses, a regulation must be found that allows private-sector use of publicly funded infrastructure.

— The establishment of the planned AI Factories should be used to promote innovative solutions from young European companies within the framework of pre-commercial procurement. One example is the use of photonic coprocessors from the company Q.ANT at the Leibniz Supercomputing Centre in Garching.

Support Ambitious AI Research and Development

— Germany and its European partners should promote research and development into AI models that offer the potential for the next fundamental breakthrough in global AI development. In light of high electricity costs, research and development into energy-efficient models should be systematically supported.

— A key obstacle to training European AI models is legally secure access to data. To this end, the General Data Protection Regulation should be amended accordingly to facilitate the training of foundation models, and opportunities for the joint training of specialized models should be created (e.g. data trustee models and the legally secure use of privacy-enhancing technologies). Another measure in the context of generative AI would be to introduce the principle of misuse, which basically allows the use of data as long as it is not misused.

— The centralized and user-friendly provision of public data, for example by federal, Länder and local administrations, must be accelerated.

— To strengthen digital sovereignty, reduce security-related dependencies on non-European AI providers and facilitate the development of derivative models, the Federal Government should work towards promoting private-sector European cooperation to develop an open-source foundation model. Since the competitiveness of such models requires iterative further development, the EU and its Member States should provide long-term support for the foundation model as anchor customers.

Strengthen AI Application in the Public Sector

— The use of AI in the public sector should be significantly expanded. To this end, the development of a Germany-wide, digitally uniform administrative infrastructure should be rapidly advanced – for example, through the Deutschland-Stack and considering European open models to ensure European digital sovereignty. A modular approach that simplifies the replacement of individual software elements in the Stack can reduce dependencies through lock-in effects.

- The extent to which the use of German and European AI solutions can contribute to strengthening digital sovereignty should be examined, even though they do not yet match the performance level of non-European providers in all fields. However, it must be ensured that the use of such solutions does not shield providers from technological competition but rather supports them in catching up with the world leaders.

Promote AI Start-ups

- Start-ups should be supported by being considered more frequently in public procurement, for example through the increased use of the special direct contract limit of €100,000 provided for in the HTAD for innovative services from start-ups.
- The issue of the taxation of employee share ownership in start-ups should be addressed much more consistently than it has been in the Future Financing Act 2023.
- The framework conditions for venture capital should be improved; in addition, the Savings and Investment Union should be promoted at EU level.
- The Federal Government should advocate for the most comprehensive 28th regime possible within the EU (cf. chapter A 4) in order to reduce the fragmentation of the Single Market and make it easier for EU start-ups to scale up.